

Functional Annotation of *purF* (Q88LD5) in *Pseudomonas putida* KT2440

Gene: *purF* (ordered locus PP_2000) · **UniProt:** Q88LD5 · **Enzyme:** Amidophosphoribosyltransferase / glutamine phosphoribosylpyrophosphate amidotransferase (GPATase, ATase) · **EC 2.4.2.14** · **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440)

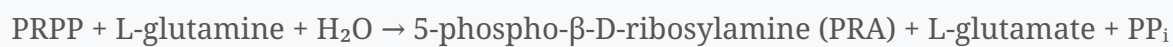
1. Summary (Answer to the Research Question)

purF encodes **amidophosphoribosyltransferase (GPATase)**, the enzyme that catalyzes the **first committed and rate-controlling step of de novo purine nucleotide biosynthesis**. In the cytoplasm it transfers the amide nitrogen of **L-glutamine** to **5-phospho- α -D-ribose-1-diphosphate (PRPP)**, producing **5-phospho- β -D-ribosylamine (PRA)**, L-glutamate, and pyrophosphate. It is a two-domain glutamine amidotransferase: an N-terminal Ntn-hydrolase glutaminase domain (catalytic Cys1) generates ammonia, which is channeled through an internal tunnel to a C-terminal phosphoribosyltransferase (PRT) domain that attaches it to PRPP. Its activity is gated by ordered substrate binding (PRPP first, which activates the glutaminase site) and feedback-inhibited by purine nucleotides (AMP/GMP/IMP), while the gene is transcriptionally repressed by the purine-responsive PurR repressor. The product PRA feeds both the purine pathway ($\rightarrow \text{IMP} \rightarrow \text{AMP/GMP}$) and thiamine biosynthesis.

Gene-identity verification: Confirmed. The gene symbol, EC number, InterPro domains (GATase_2, Ntn-hydrolase N, PRTase), organism, and the direct Q88LD5 sequence (501 aa beginning **M-C-G-I-V-G**, the canonical GPATase N-terminus; 65% identical to experimentally characterized *E. coli* PurF) all agree. This is a genuine, well-characterized amidophosphoribosyltransferase, not an ambiguous or mis-annotated symbol.

2. Primary Function: Reaction and Substrate Specificity

Reaction (EC 2.4.2.14):



This is the entry reaction of de novo purine biosynthesis. In *Salmonella/E. coli* genetics, "Glutamine phosphoribosyl pyrophosphate (PRPP) amidotransferase is the product of the *purF* gene ... and catalyzes the synthesis of PRA from PRPP and glutamine. Strains lacking PurF require exogenous addition of purines for growth" (Koenigsknecht et al., 2012,).

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The enzyme is also confirmed as a distinct, competitively inhibitable target: feruloyl amide "is a competitive inhibitor of glutamine PRPP amidotransferase (PurF)" (Pisithkul et al., 2015,).

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Substrate specificity and kinetics. The two substrates are the phosphoribosyl donor **PRPP** and the amide-nitrogen donor **L-glutamine** (free ammonia can substitute at high concentration in vitro, but glutamine is the physiological donor). Kinetic analysis of the mechanistically conserved enzyme gave dissociation constants of **80 μM for PRPP** and **420 μM for L-glutamine**, "with P-Rib-PP bound first with positive cooperativity for interaction with a second site on the catalytically active dimer" (Schoettle et al., 1997,).

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Thus catalysis follows an **ordered mechanism in which PRPP binds before glutamine**.

3. Catalytic Mechanism and Domain Architecture

GPATase is a **bifunctional, two-domain enzyme**:

- **N-terminal glutaminase domain** — a class-II (Ntn) glutamine amidotransferase (InterPro GATase_2, IPR017932; Ntn-hydrolase N, IPR029055). It belongs to the **N-terminal nucleophile (Ntn) hydrolase superfamily**; the mature enzyme is generated by autocatalytic removal of an N-terminal propeptide to expose the catalytic **Cys1**, and "propeptide cleavage was strongly inhibited by replacement of the cysteine nucleophile and two residues of an oxyanion hole that are required for glutaminase function" (Li et al., 1999,).

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- **C-terminal phosphoribosyltransferase (PRT) domain** — a type-I PRTase fold (IPR000836/IPR029057) that binds PRPP and receives the ammonia.

Active-site chemistry (from *E. coli* crystallography + mutagenesis, Kim et al., 1996,

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the free α -amino group of Cys1 acts as the proton acceptor/donor; Asn101 and the Gly102 backbone form the oxyanion hole; Arg73 and Asp127 bind glutamine; and Tyr74 couples the glutamine and PRPP sites.

Substrate-gated activation. The basal PRPP-independent glutaminase activity is only ~0.3% of the fully active enzyme; "Binding of PRPP activates the enzyme by a structural change that lowers the K_m for glutamine 100-fold and couples glutamine hydrolysis to synthesis of 5-phosphoribosylamine"

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The ring oxygen of PRPP is required for this activating conformational change (Kim et al., 1995,

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This structural gating ensures glutamine is not wastefully hydrolyzed unless the acceptor PRPP is present — the mechanistic counterpart of the PRPP-first ordered kinetics.

Intramolecular ammonia channeling. The ammonia generated at the glutaminase site is not released to solvent; GPATase "catalyzes the synthesis of 5'-phosphoribosylamine in a reaction that involves the translocation of ammonia along an intramolecular tunnel linking the two active sites," behaving as a "pipe" through which ammonia travels; the L415A mutation reduces channel efficiency (Wang et al., 2009,

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4. Localization

GPATase is a **soluble cytoplasmic enzyme**. All structurally characterized bacterial orthologs (*E. coli*, *B. subtilis*) are cytosolic proteins crystallized as soluble oligomers

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8663035), and the *P. putida* sequence has no signal peptide or transmembrane segment. It

therefore carries out its function **in the cytoplasm**, where PRPP and glutamine are supplied by central metabolism. (The mammalian cytosolic "purinosome" metabolon is a eukaryote-specific organization and is not expected in bacteria; notably, even in humans amidophosphoribosyltransferase is the one DNPB enzyme that does not stably join the PAICS-centered purinosome interactions — He et al., 2022, [P 35331738](#)).

5. Pathway Context and Biological Process

PurF initiates the **ten-step de novo purine pathway**: $\text{purF} \rightarrow \text{purD} \rightarrow \text{purN/purT} \rightarrow \text{purL} \rightarrow \text{purM} \rightarrow \text{purK/purE} \rightarrow \text{purC} \rightarrow \text{purB} \rightarrow \text{purH}$, yielding **IMP**, from which **AMP** and **GMP** are made. Its product PRA is also the branch-point precursor of **thiamine (vitamin B1)** biosynthesis: "Phosphoribosylamine (PRA) is an intermediate in the biosynthetic pathway that is common to thiamine and purines" ([P 23133571](#)).

By consuming PRPP and glutamine, PurF sits at the intersection of nucleotide supply, one-carbon/nitrogen metabolism, and central carbon metabolism.

6. Regulation

PurF flux is controlled at **two layers**:

1. **Allosteric feedback inhibition (fast, enzyme level)**. End-product purine nucleotides inhibit the enzyme. In the *E. coli* structure the feedback inhibitor **AMP binds the PRPP catalytic site**, locking the inhibited conformation with the two active sites disconnected (Muchmore et al., 1998, [P 9514258](#));

AMP is **competitive with respect to PRPP** ($K_d \sim 40 \mu\text{M}$), and IMP and GMP also inhibit ([P 9174353](#)).

A separate allosteric nucleotide site also exists

([P 7542237](#)).

2. **Transcriptional repression (slow, gene level).** In *E. coli* the LacI-family repressor **PurR**, with a purine corepressor (hypoxanthine/guanine), binds a high-affinity operator at *purF* to repress de novo purine/pyrimidine genes (Schumacher et al., 1994,

[P 8089849](#));

Devroede et al., 2004,

[P 14741201](#)).

In *B. subtilis* a non-homologous PurR senses excess adenine via the PRPP pool (Weng et al., 1995,

[P 7638212](#)).

P. putida KT2440 encodes a LacI-family PurR ortholog, so analogous purine-responsive repression of *purF* is expected.

7. Evolutionary / Structural Classification

Bacterial GPATases form two subfamilies: "The *E. coli* enzyme is a prototype for the **metal-free** GPATases, whereas the *B. subtilis* enzyme represents the **metal-containing** enzymes ... a common ancestor ... may have included an Fe-S cluster"

([P 9514258](#)).

The *B. subtilis* [4Fe-4S] cluster is remote from the active sites and confers oxygen-triggered turnover rather than catalysis. **Q88LD5 is a gammaproteobacterial, E. coli-like, metal-free GPATase** (HAMAP rule MF_01931), so it is not expected to be intrinsically O₂-sensitive via a metal center — consistent with *P. putida*'s aerobic lifestyle.

Direct evidence for the *P. putida* protein: Q88LD5 is 501 aa, begins **MCGIVGIVG** (conserved Cys1 nucleophile motif), and is **65.1% identical to E. coli PurF (P0AG16)** over the full length (Needleman-Wunsch, this work). Crucially, **all six experimentally-defined catalytic residues are perfectly conserved** in Q88LD5 by alignment (this work): the nucleophile **Cys1**, glutamine-binding **Arg73** and **Asp127**, inter-site coupling **Tyr74**, and oxyanion-hole **Asn101/Gly102** (residues defined in *E. coli* by Kim et al.,

).

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This residue-level conservation justifies transferring the *E. coli* experimental mechanism directly to the *P. putida* ortholog.

8. Supported and Refuted Hypotheses

Supported: - H1: *purF* encodes amidophosphoribosyltransferase (EC 2.4.2.14) catalyzing PRPP + Gln → PRA. ✓ (PMIDs 23133571, 26070680; sequence) - H2: Two-domain Ntn-hydrolase mechanism with Cys1 nucleophile and PRPP-gated glutaminase. ✓ (PMIDs 8663035, 10049369, 7542237) - H3: Intramolecular ammonia channeling between active sites. ✓

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- H4: Ordered kinetics (PRPP first); feedback inhibition by AMP/GMP/IMP. ✓ (PMIDs 9174353, 9514258) - H5: Dual regulation — allosteric + PurR transcriptional repression. ✓ (PMIDs 8089849, 14741201, 7638212) - H6: Cytoplasmic localization; entry to purine + thiamine pathways. ✓ (PMIDs 23133571, 9514258) - H7: *P. putida* enzyme is a metal-free, *E. coli*-type GPATase. ✓

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65% identity)

Refuted / not applicable: - The bacterial enzyme does not organize into a mammalian-style purinosome, and (unlike the *B. subtilis* enzyme) is not predicted to carry an oxygen-sensitive Fe-S cluster.

9. Limitations and Future Directions

- No *P. putida*-specific enzymological or structural study of PurF was found; mechanistic detail is transferred by strong orthology (65% identity, conserved catalytic residues) from *E. coli*/*B. subtilis*, which is well justified but not organism-specific proof.
- Direct confirmation of PurR-mediated *purF* regulation and the exact operon structure in *P. putida* KT2440 would benefit from ChIP/RNA-seq or reporter data.

- A crystal or cryo-EM structure of the *P. putida* enzyme would confirm the metal-free assignment and quaternary state directly.

Key

references:

P 23133571

P 26070680

P 8663035

P 10049369

P 7542237

P 19921932

P 9174353

P 9514258

P 8089849

P 14741201

P 7638212

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Sequence analyses (Q88LD5 vs P0AG16) performed in this study.